

Data Cleaning Report

Sales Performance Analytics Project

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Introduction

This report documents the data cleaning process performed on an e-commerce sales dataset that contains 1,000 records and 13 variables. The objective was to prepare the data for exploratory analysis into which factors that drive sales across products, cities, and time periods. Data cleaning was necessary because missing values can affect calculations, visualizations, and overall data quality, leading to inaccurate analytical results.

Dataset Overview

Item	Value
Number of Records	1,000
Number of Variables	13
Source	Kaggle
File Type	CSV

Tools Used

Tool	Purpose
Excel	Initial data inspection, missing value inspection, and data cleaning.
Excel (PivotTables)	Data analysis
Tableau	Data visualization

Issues Identified

- Missing values
 - Found 201 missing values in review_score
 - Found 103 missing values in gender

Data Cleaning Procedures

- The Find and Replace function in Excel was used to fill in the missing values:
 - Missing values in review_score were replaced with a 0 to indicate that no review score was available.
 - Missing values in gender were replaced with “Unknown” rather than assigning a specific gender. This distinguishes missing information from valid gender categories.

Summary of Changes

- There was a total of 304 missing values:
 - 201 values in review_score
 - 103 values in gender
- Missing values in review_score were replaced with a 0, while missing values in gender were replaced with “Unknown.”
- The cleaned dataset still has 1,000 records and 13 variables.

Limitations

While these changes fixed the issues with missing values, it may not truly represent customers' true gender or what they left for a review score. A customer's gender could either be something that is not male or female, or they simply did not disclose their gender. As for review_scores, customers could have left a review score of 0 or did not leave a review at all.

Conclusion

- The dataset was successfully cleaned and prepared for analysis.
- Missing values were identified and replaced.
- The resulting dataset contains 1,000 records with all missing values addressed. By replacing missing review scores and gender values with standardized entries, the dataset is more consistent and suitable for exploratory analysis and visualization.